Tsunamis

Earthquakes happen on the ocean floor as well as on land. The shock waves ripple up and out through the water, pushing up enormous waves called tsunamis. Out on the open ocean, they can be quite hard to detect because they are very broad, but when they sweep into shallow water they get much steeper and more destructive.

SUBMARINE EARTHQUAKE

When the ocean floor plows beneath another slab of Earth's crust, the moving plates tend to lock together, build up strain, then snap apart to cause ocean-floor earthquakes. As the upper plate springs back up, it pushes the water above it up into a heap. This huge ripple then surges across the ocean at high speed as a series of giant waves that can cause chaos when they strike land.

▸ MONSTER WAVES
If tsunamis drive into shallow water, they slow down due to friction with the seabed, and get much steeper. As they surge ashore, they overwhelm anything in their path.

EXPLODING ISLANDS

Tsunamis can be generated by massive landslides that plunge into the sea and push up giant waves. A volcanic island can also trigger a tsunami if it erupts so violently that it blows itself apart. Seawater cascading into the red-hot heart of the shattered island boils instantly and explodes, with a shock wave that turns into a tsunami. This can be just as lethal as the eruption itself.

◂ VOLCANIC BOMB
In 1883, the volcanic island of Krakatoa near Java exploded, causing a deadly tsunami. The dark island in this image is a new volcano erupting from the flooded crater of the old one.